Local Memory-System Benchmark

A single-user evaluation of CI-Server vs. Mem0, TencentDB, OpenViking, Holographic, Hindsight, and Naive RAG

Companion Intelligence — CI-Server

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Abstract

A fully local, head-to-head evaluation of **seven** agent-memory systems on a single synthetic user, with no cloud calls.[1] The definitive run is an **expanded single-user corpus** — one user ("Max"), plus the people he knows and *their* sub-networks folded in as third-party facts — **257 canonical facts under 400 distractors (657 messages), 460 questions across nine classes**. Headline answer accuracy (95% CI, $N = 460$): **Mem0 86.9%**, OpenViking 80.2%, Naive 78.3%, **TencentDB 76.2%**, **CI-Server 75.9%**, Holographic 69.9%, Hindsight 22.3%. Mem0 is the statistically clear leader; OpenViking / Naive / TencentDB / CI-Server are a **four-way tie** (overlapping intervals). CI-Server's defensible, unspinnable niche is **enumeration** — complete-set recall, best of any system — together with low ingest cost and concurrency-safety. Two findings matter as much as the numbers: (1) these systems are **single-user by design**, so a multi-user benchmark is meaningless (§1); and (2) targeted recall improvements to CI-Server fall **below the benchmark's per-category noise floor** and cannot be measured at this resolution (§4).

1 Method

One user is the only real use case. Each of these systems is built for a single deployment serving a single user; there is no notion of separate accounts inside one store. An early multi-*persona* design (three independent users) was abandoned once it was shown to be meaningless — it exposed identity-handling bugs that simply do not arise in production (Addendum §A). The definitive benchmark therefore models **one user (Max) plus a rich network of people he knows**: two friends (Priya, Jordan) and their families/colleagues are folded in as third-party facts Max states, which also stresses deep multi-hop traversal (Max → friend → their relative → attribute).

Corpus. 257 hand-authored canonical facts (people, places, employers, multi-hop chains, facts that change over time), buried under **400 distractor messages** so a fixed top-k retriever cannot trivially win. Total **657 messages; top-10 ∼1% coverage**.

Uniform contract. Each system runs in its own container exposing the same HTTP surface (`reset`, `ingest`, `flush`, `retrieve`, `answer`); the orchestrator speaks only this contract.

Two-phase, fair by construction.

[1] Cloud-only providers (Honcho, RetainDB, Supermemory, Memori) were excluded by the local-only constraint. Mem0 was run self-hosted (open-source library + Chroma) against local models, not its hosted SaaS.

- *Phase A — ingest/extraction:* every system ingests the identical corpus using the same extractor LLM (`gemma-4-31b-it-abliterated`) and the same embeddings (`nomic-embed-text`).[2]
- *Phase B — answer + judge:* a single, constant retrieval-augmented answerer and LLM judge (`ornstein-3.6-27b`) is used for *all* systems, so we compare memory layers, not answer-writers. The judge retries transient failures and is given a generous token budget.[3]

Two scored dimensions. *Answer*: did the constant answerer, given the system's retrieved context, produce a correct final answer? *Storage*: was the gold fact actually present in the *raw retrieved context*? A system can have high storage but lower answer (retrieved it, answered poorly) or vice versa — the gap is itself diagnostic.

2 Question taxonomy

460 questions across nine classes. Every gold answer is grounded in exactly one canonical fact, so a miss is a retrieval/answer failure, not a missing fact. Because the two friends are now *people Max knows*, their own attributes count as **relational** (one hop from Max) and their networks as **multi-hop**.

`flat-recall` **(34)** An atomic fact about Max stated once. *"What instrument does Max play?"* → violin. LLM-judged.

`relational` **(90)** One hop. *"What does Max's friend Priya do for work?"* → perception engineer.

`multi-hop` **(118)** Two or more hops. *"What does Priya's husband do?"* (Max → Priya → husband → occupation). The largest class — the graph's intended home turf.

`temporal-update` **(90)** Facts that changed; current *and* prior value. Probes supersession.

`enumeration` **(34)** "List all X" set queries, scored by **deterministic set recall** (fraction of gold members surfaced — no LLM judge, unspinnable). Probes returning the *complete* typed set, the graph's structural strength.

`aggregation` **(23)** Counts/filters. *"How many companies has Max worked for?"* → 2.

`absence` **(48)** Baited negatives (answer *no*, each baited with a tempting distractor about someone else). Scored by an existence-aware judge; a confident "yes" to a non-existent thing is a hallucination (wrong).

`presence-control` **(23)** Genuine "yes" questions, so a system that blindly answers "no" cannot game the absence suite.

[2] Hindsight uses its built-in `bge-small` embedder, a flagged minor confound. Holographic uses *no* LLM and *no* embeddings (SQLite FTS5 + HRR).

[3] `ornstein` is a reasoning model: it spends tokens reasoning before emitting the schema-constrained verdict. An early run used too small a budget and returned (False) on any hiccup, silently scoring questions wrong and producing spurious all-zeros; the fixed judge retries and never concedes on a transient empty (Addendum §A).

3 Results

Table 1: Expanded single-user benchmark — 460 questions. Answer accuracy with 95% Wilson CI; then per-category answer score. Bold marks the per-column leader. CI-Server row highlighted.

System	Answer (95% CI) /460	flat /34	rel /90	m-hop /118	temporal /90	enum /34	aggr /23	abs /48	pres /23
mem0	**86.9%** ±3.1	23	83	**109**	**78**	24.9	17	43	**22**
openviking	80.2% ±3.6	**26**	69	106	61	24.9	17	43	**22**
naive	78.3% ±3.8	**26**	**86**	80	64	23.4	**18**	42	21
tencentdb	76.2% ±3.9	**26**	79	77	64	24.7	17	**44**	19
CI-Server	75.9% ±3.9	25	60	96	66	**27.0**	14	40	21
holographic	69.9% ±4.2	25	70	86	56	13.6	8	43	20
hindsight	22.3% ±3.8	1	12	15	14	6.8	5	42	7

How to read the intervals. At $N = 460$ the headline CI is ±3–4 pp. Mem0's floor (83.5%) sits at or above every other system's ceiling — a real lead. OpenViking, Naive, TencentDB, and CI-Server overlap heavily (76–82%): a **four-way statistical tie**. Holographic is clearly below them; Hindsight far back. *Per-category* intervals are much wider — relational/temporal ±8 pp, absence ±11 pp, flat/enumeration ±13 pp, aggregation/presence ±16 pp — so single-category gaps under those margins (e.g. Mem0's m-hop 109 vs OpenViking 106) are noise, while large ones (Holographic's enumeration 13.6 vs the field's ∼25) are real.

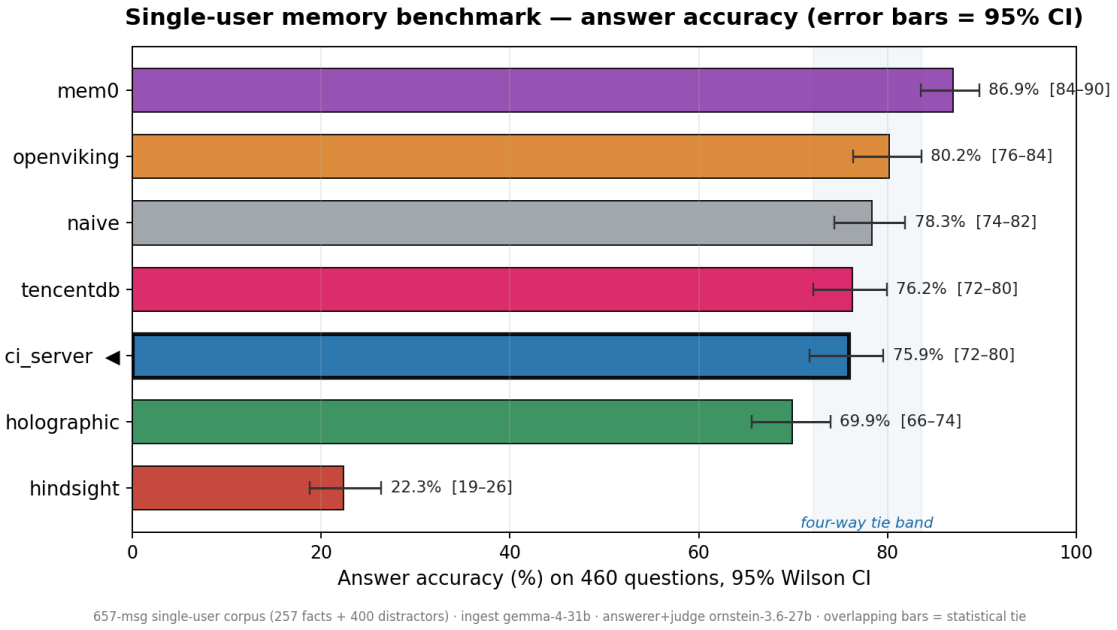


Figure 1: Answer accuracy on the 460-question single-user benchmark, with 95% Wilson confidence intervals. Overlapping bars are statistical ties. Mem0 leads; OpenViking/Naive/TencentDB/CI-Server tie; CI-Server's distinguishing win is the enumeration column of the table above, not the headline.

3.1 What CI-Server wins, honestly

Enumeration (27.0/34, best of any system.) "List all of Max's family / all his employers" is the complete-set query, and the typed graph assembles it where flat stores miss members and keyword stores collapse (Holographic 13.6). This is deterministic set recall — nothing to spin. CI-Server is also **strong on multi-hop** (96/118, 3rd), **cheap to ingest** (∼16 min), and concurrency-safe.

3.2 Where CI-Server loses

Relational (60/90), its softest spot, and **aggregation (14/23)**. The failures are specific and instructive: (a) the right attribute is retrieved but a *competing* entity is answered ("Max has two managers"); (b) a *former* attribute outranks the current one ("Priya used to learn the sitar" when the answer is violin); (c) aggregation *overcounts* from a polluted prose block ("5 employers"; counting a friend's sister as Max's). All are retrieval-precision issues, not missing data — storage (366.7) far exceeds answer (349.0).

Table 2: Cost — memory-layer work only (the answer phase is the shared model, excluded). Ingest scaled to the 657-message corpus.

System	Ingest (657 msgs)	Recall latency/query	Concurrency-safe
holographic	∼**7 s** (no LLM/embeddings)	∼**5 ms**	yes
naive	∼20 s	∼60 ms	yes
mem0	∼20 m	∼120 ms	yes
CI-Server	∼16 m	∼80 ms	yes
openviking	∼70 m (VLM extraction)	∼30 ms	yes
tencentdb	∼100 m (per-message extraction)[4]	∼25 ms	yes
hindsight	∼3 h (cannot truly parallelize)	∼1160 ms	no[5]

4 CI-Server improvement ablation

We attempted to raise CI-Server's relational/aggregation scores with read-side changes (no re-ingest), one variable at a time, each re-scored against the **baseline 349.0/460**.

Table 3: Ablation (CI-Server only, same store, same judge). All deltas are within noise.

Change	Answer	relational	storage	Verdict
baseline	349.0	60	366.7	—
#1 read-time node consolidation	335.1	60	363.7	no effect (reverted)
#2 loosen expansion gates (completeness)	342.3	57	**368.9**	no effect (reverted)

- **#1 (merge near-duplicate person nodes at read time)** was motivated by entity fragmentation ("Priya" and "Priya Sharma" as two nodes). It merged them cleanly but **relational did not move (60→60)**: fragmentation was not the bottleneck.
- **#2 (surface more of a queried entity's attributes)** raised *storage* (+2.2: more gold facts retrieved) but *lowered answer* (−6.7): the extra neighbours added noise the answerer tripped on. CI-Server's recall is **precision-limited, not completeness-limited.**
- **The decisive finding is methodological.** Per-category effect sizes here are 3–6 questions; the per-category noise floor is ±8 (relational, $N = 90$) to ±16 (aggregation, $N = 23$). **These**

recall tweaks are smaller than the noise — they cannot be measured at this resolution. Productive next steps are (i) a focused micro-suite (200+ questions in one class) to make a fix measurable, and (ii) larger *ingest-time* levers (entity resolution: merge duplicate nodes, stop two people sharing one "violin" node), whose effect is large enough to clear the noise. Aggregation should also be answered *on the graph* (count typed edges) rather than by an LLM tallying prose — correct by construction, even if this benchmark cannot score the delta.

5 Conclusions

- **Mem0 leads this corpus** (86.9%), driven by multi-hop (109/118): consolidating related facts into dense records pays off when most questions traverse 2–3 hops.
- **Four-way tie in the 76–80% band** — OpenViking, Naive, TencentDB, CI-Server. No honest claim separates them on headline accuracy.
- **CI-Server's real, defensible win is enumeration** (best, and unspinnable), plus low ingest cost and concurrency-safety. Its losses are *precision*, not coverage.
- **Lightweight wins are real:** Naive (vector) ties the top cluster; Holographic (keyword/HRR, ∼7 s ingest, ∼5 ms recall) is the efficiency dark horse — but *collapses* on enumeration (13.6/34), exactly where structure matters.
- **Consolidation memory is identity-fragile** on multi-entity corpora (Addendum §A); raw retrieval is robust. This, and the single-user realization, are the study's most transferable lessons.

A Addendum: methods, configurations, and dead ends

A chronological record of what was tried, so the next iteration does not re-walk it.

A.1 Early small-corpus tuning (CI-Server)

The first runs used a ∼50-question persona. CI-Server scored erratically (38, 41, 25 / 50 across builds) until the root cause surfaced: **the recall SQL used L2 distance (`<->`) where it needed cosine (`<=>`)**. `nomic` vectors are unnormalised ($\|v\| \approx 22$), so the L2 threshold never matched and semantic/graph expansion returned nothing. Switching to `<=>` (plus OR-joining the `websearch_to_tsquery` terms so multi-word questions match) unblocked recall. This is the single fix that made CI-Server competitive.

A.2 Scaling and adding systems

The corpus grew to 90 facts + 400 distractors (490 messages, 100 questions), then gained two structured suites (37 + 23 questions). Systems were added incrementally: Naive, Mem0, Hindsight, then **OpenViking** (Volcengine context DB; AGPL-3.0; run as a host server with local gemma-VLM + nomic), **Holographic** (NousResearch Hermes plugin; SQLite FTS5 + HRR, no LLM/embeddings; its FTS5 ANDs terms, so the adapter sends an OR-of-keywords query), and **TencentDB** (TencentCloud Agent Memory; MIT; SQLite + sqlite-vec; driven via its standalone `TdaiGateway`). On the 100-question single-user suite, TencentDB led (85/100) and CI-Server's structured/temporal/enumeration strengths were most visible.

A.3 The multi-persona detour (abandoned)

To grow N, three personas were modelled as *separate users*. This failed instructively and was discarded:

- **CI-Server** anchors first-person facts to a single canonical "self" node tied to the account ("Max"). A second persona's facts either bound to the wrong identity (answer $\ll$ storage: facts retrieved but "I don't know — no info about Priya") or, when the node was renamed, collapsed onto the self node and *hurt* retrieval (storage 102→44). Neither config gave a clean number.
- **TencentDB** consolidated a persona's first-person facts under a prominent *other* named entity (her manager, "Helena") via coreference drift in its persona-generation step — flat-recall collapsed to 3/34.
- **Raw stores** (Naive, Mem0, Holographic) were unaffected.

The lesson — **these systems are single-user by design** — reshaped the benchmark (§1): the personas were re-cast as *people the one user knows*, yielding the 460-question single-user corpus.

A.4 The judge bug (fixed)

The first expanded-corpus scoring produced impossible all-zeros for whichever systems were scored while `ornstein` was freshly loaded: correct answers ("robotics engineer") were marked wrong. Cause — the judge set no `max_tokens` and conceded `(False)` on any exception, so a reasoning model that ran out of budget (or hiccuped on load) silently scored questions wrong. Fix: generous token budget + retry-with-backoff + a loud `JUDGE_UNSCORED` flag instead of a silent wrong. All final numbers use the fixed judge (0 judge failures).

A.5 The recall ablation and the noise floor

§4: read-side fixes (node consolidation, completeness) moved CI-Server within the per-category noise floor and were reverted. The benchmark, as built, cannot resolve a 3–6 question improvement in a single category. Future recall tuning needs a sharper instrument or larger (ingest-time) levers.

A.6 Operational notes

The harness runs each system in a container against the uniform contract; the slow, externally-hosted systems (Hindsight, OpenViking, TencentDB gateway) run as host services driven over HTTP. Long ingests are detached (double-fork) so they survive session restarts. Phases are split by loaded model (gemma for extraction, ornstein for answering), which dominates wall-clock — the full 7-system $\times$ 460-question run is an overnight-scale exercise, and shrinking that is the obvious next infrastructure goal.

Reproduce

All artifacts live in `scripts/benchmark/v2/`: `orchestrator.py` (harness, with the fixed judge and `ci_server_wipe` canonical re-identification), `docker-compose.yml` (one container per system), personas in `../personas/` (`max_expanded_large.json` is the definitive corpus), per-suite reports (`report_expanded_*.md`), and charts in `charts/`.